# EE133/233 FM Radio Receiver

## System Overview — Final Project

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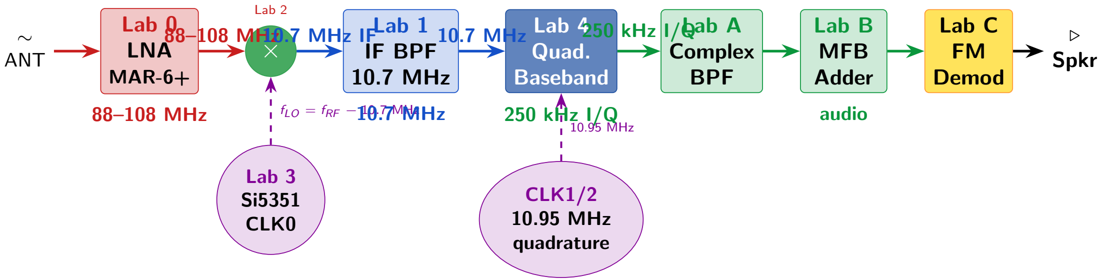
EE133/233 — Stanford University

March 2026

# Full System Architecture




## RF Stage

Antenna → LNA → Mixer
88–108 MHz → 10.7 MHz

## IF Stage

IF BPF selects 10.7 MHz
Image rejection + channel filtering

## Baseband

I/Q → Complex BPF → FM demod
250 kHz → audio

# Lab 0 — MAR-6+ Low-Noise Amplifier

## What it does

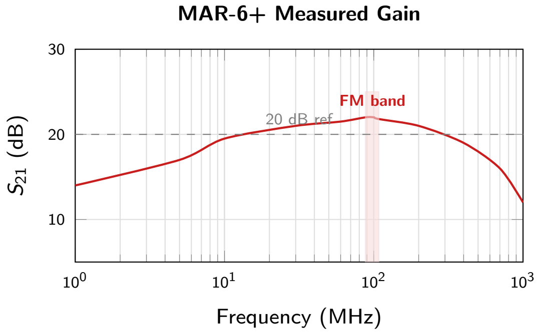
The MAR-6+ is the first stage amplifier. It sets the noise floor for the entire receiver (Friis equation):

$$F_{sys} \approx F_1 + \frac{F_2 - 1}{G_1}$$

With $G_1 \approx 22$ dB, noise from all later stages is suppressed by $\times 158$.

| Parameter | Value |
|---|---|
| IC | MAR-6+ (Mini-Circuits) |
| Gain | $\approx 22$ dB @ 100 MHz |
| NF | 2.3 dB |
| Bandwidth | 1 MHz – 1 GHz |

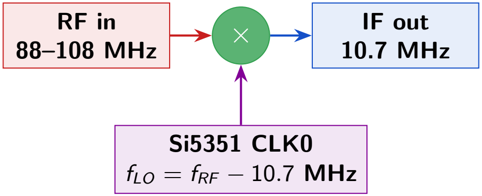
## $S_{21}$ gain vs frequency (NanoVNA measurement)




−48 dBm

# Lab 2 — ADE-1 Diode Ring Mixer: First Downconversion

## Frequency translation

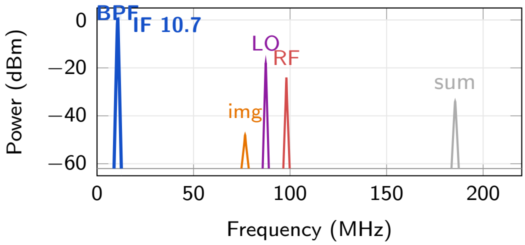



**Example** (tuned to 98.1 MHz):

$$f_{LO} = 98.1 - 10.7 = \mathbf{87.4}\ \text{MHz}$$

$$f_{IF} = |f_{RF} - f_{LO}| = \mathbf{10.7}\ \text{MHz}$$

## Mixer output spectrum (RF = 98.1 MHz)




| Product | Formula | Value |
|---|---|---|
| **IF** | $\lvert f_{RF} - f_{LO} \rvert$ | **10.7 MHz** |
| Image | $f_{LO} - 10.7$ | 76.7 MHz |
| LO leak | $f_{LO}$ | 87.4 MHz |
| Sum | $f_{RF} + f_{LO}$ | 185.5 MHz |

# Lab 3 — Si5351 LO: Quadrature Generation

### Silicon Labs Si5351 Fractional-N PLL

- Programmable PLL controlled via $I^2C$
- Controlled by **Raspberry Pi Pico 2W** over WiFi HTTP
- **CLK0:** first LO for Lab 2 mixer
  $f_{CLK0} = f_{RF} - 10.7$ MHz
  Covers 77.3–97.3 MHz (full FM band)
- **CLK1/CLK2:** quadrature LO at 10.95 MHz
  90° offset via `set_phase(CLK2, 64)`
  VCO runs at $64 \times f_{out}$ on PLLA

### Quadrature clock generation (Pico 2W firmware):

```
// CLK1 = 0°, CLK2 = 90° (both on PLLA)
si5351.set_freq(quad_hz * RATIO, PLLA, CLK1);
si5351.set_freq(quad_hz * RATIO, PLLA, CLK2);

si5351.set_phase(CLK1, 0);  // 0°
si5351.set_phase(CLK2, 64);  // 90°
si5351.pll_reset(PLLA);     // lock phase
```

# Lab 3 — Station Tuning and Phase Control

**Station tuning via HTTP API:**

```
// First LO --- tune to FM station X:
  GET /api/clk2/set?mhz=X
  CLK0 = X - 10.700 MHz

// Quadrature LO --- fixed at 10.95 MHz:
  GET /api/quad/set?mhz=10.95
```

**Why phase matters:**

- `pll_reset(PLLA)` enforces coherent start
- RATIO $= 64 \Rightarrow$ VCO at $64 \times 10.95 = 700.8$ MHz
- One VCO cycle $= \frac{1}{700.8\text{ MHz}}$; 64 cycles $= \frac{1}{10.95\text{ MHz}} =$ one full period $\Rightarrow$ **exactly 90°**

# Lab 1 — 10.7 MHz IF BPF & Impedance Matching

**The problem**

Ceramic BPF impedance $Z_0 \approx 330\,\Omega$ vs system $50\,\Omega$.

**Without matching:**

- VSWR $= 6.8$ $\quad S_{21} \approx -20$ dB
- Most power reflected; signal lost

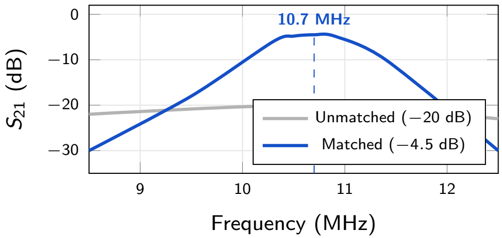
**L-network solution** ($50 \to 330\,\Omega$)**:**

$$L_s = 1.76\,\mu\text{H}, \quad C_p = 106.7\,\text{pF}, \quad Q_T = 2.37$$

**With matching:**

- VSWR $\approx 1.05$ $\quad S_{21} \approx -4.5$ dB
- **15.5 dB improvement**

## $S_{21}$: matched vs unmatched



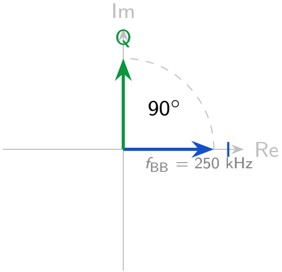
# Lab 4 — Quadrature I/Q Downconversion

**Three cascaded boards:**

1. **ADT1-4W** transformer: 1 : 4 impedance, SE $\rightarrow$ differential
2. **74CBTLV3253**: quadrature analog switches
   Driven at $0^\circ/90^\circ/180^\circ/270^\circ$ by Si5351 CLK1/CLK2
3. **AD4940-1** FDA: dual LPF $+$ gain $= 2$

*Note: 74ALVC74 (÷4) not used. Si5351 provides quadrature directly via* `set_phase(CLK2,64)`.

**I/Q phasor view**



# Lab 4 — Quadrature I/Q Downconversion

**Frequency math:**

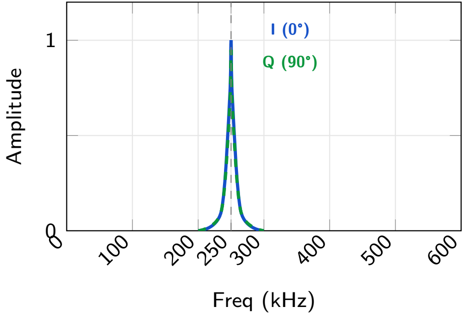
$$f_{\text{quad}} = 10.95 \text{ MHz}$$

$$f_{\text{low-IF}} = |f_{\text{IF}} - f_{\text{quad}}| = |10.70 - 10.95| = \mathbf{250} \text{ kHz}$$

**Measured result:**

I and Q outputs both show tone at 250 kHz.
C1: 5.45 $V_{\text{pk-pk}}$, C2: 5.46 $V_{\text{pk-pk}}$ $\Rightarrow$ balanced amplitudes confirm $90^\circ$ phase relationship.

**Spectrum at baseband:**

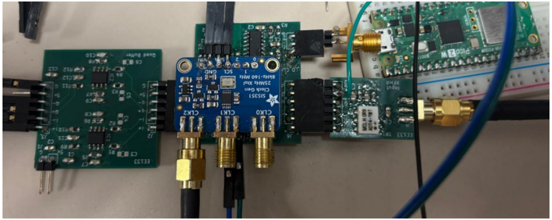


# Experiment 1 — Quadrature + LPF: Hardware Validation

**Setup: ADT1-4W + 74CBTLV3253 + AD4940-1 LPF (no complex BPF)**

IF $= 10.7$ MHz $\quad f_{\text{quad}} = 10.95$ MHz $\quad \Rightarrow$ low-IF $=$ **250 kHz**

## Oscilloscope (I/Q waveforms)



C1 (orange) = I channel  C2 (blue) = Q channel

Both at 250 kHz, balanced amplitudes $\approx 5.45$ $\text{V}_{\text{pk-pk}}$

## Spectrogram (Digilent WaveForms)

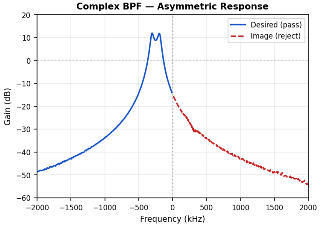

Tone at 250 kHz clearly visible in both channels.

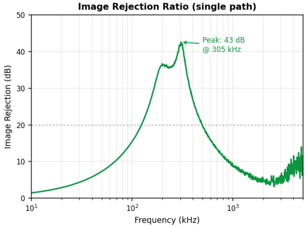
Both I and Q pass the desired tone — image rejection not yet applied.

# Lab A — Complex BPF: Simulation Results

## Single-path measurement (polyproc.m, esI/esQ)

Asymmetric response: desired pass vs image reject, + image rejection ratio:



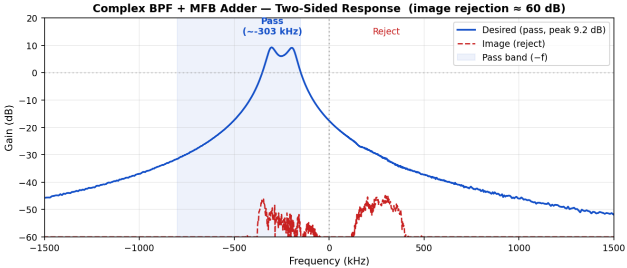

**Lab A** — Cross-coupled Tow-Thomas (4× THS4551)

- Cross-coupled I/Q paths ⇒ **asymmetric** frequency response
- Passes −300 kHz sideband (desired)
- Rejects +300 kHz sideband (image)
- Single path: ≈ **43** dB rejection

# Lab B — MFB Adder: Simulation Results

## Full chain (ActProcB.m, espI/Q + esmI/Q)

Complex BPF + MFB adder two-sided response:




## Lab B — MFB Adder (1× THS4551)

- Combines $I' + Q'$ into one real differential signal
- Full chain rejection: $\approx$ **60** dB at 303 kHz
- Output feeds Lab C FM demodulator

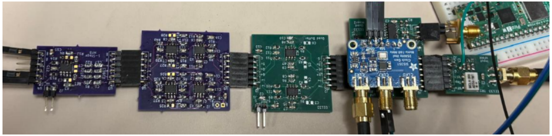
| Path | Image Rej. |
|---|---|
| Single path | $\approx$ 43 dB |
| Full chain | $\approx$ **60 dB** |

# Experiment 2 — Quad + Complex BPF + Adder: Hardware Validation

## Setup: Exp. 1 chain + Complex BPF (Tow-Thomas, 250 kHz) + MFB Adder

Expected: desired tone passes, image tone **suppressed by $\approx$ 60 dB**

### Oscilloscope (after Complex BPF + Adder)



C1 (orange): tone at 250 kHz (desired) — present

C2 (blue): image sideband — **strongly suppressed**

Desired tone at 250 kHz retained.

Image component greatly attenuated — confirms $\approx$ 60 dB rejection.

### Spectrogram (after Complex BPF + Adder)

**Compared to Exp. 1:** C2 now shows image suppression instead of equal amplitude.

**Lab C — FM Demodulation Board**

- Inputs: In$+$, In$-$, VDD, GND (differential)
- FM discriminator: $\Delta f$ deviation $\rightarrow$ voltage
- Audio band recovered: 20 Hz – 15 kHz $\rightarrow$ Speaker

**Key measurements:**

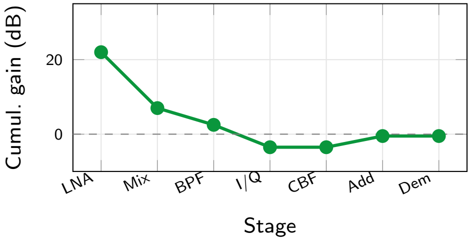
- **LNA:** $+22$ dB gain, 2.3 dB NF (FM band, flat)
- **IF BPF:** $+15.5$ dB after L-network matching
- **Quadrature:** 250 kHz low-IF, balanced I/Q confirmed
- **Complex BPF:** $\approx$ 60 dB image rejection
- **Audio:** recovered at speaker ✓

## Conclusion — End-to-End FM Receiver

**One station's journey (98.1 MHz):**

| Stage | In | Out |
|---|---|---|
| Lab 0 LNA | $-70$ dBm RF | $-48$ dBm RF |
| Lab 2 Mixer | $98.1 + 87.4$ MHz | 10.7 MHz IF |
| Lab 1 BPF | mixed products | 10.7 MHz clean |
| Lab 4 I/Q | 10.7 MHz IF | 250 kHz I/Q |
| Lab A CBF | $\pm$250 kHz | $-250$ kHz only |
| Lab B Adder | $I'$, $Q'$ | combined real |
| Lab C Demod | FM signal | audio |

**Gain budget across chain:**




**Antenna $\longrightarrow$ Speaker**

88–108 MHz $\rightarrow$ 10.7 MHz $\rightarrow$ 250 kHz $\rightarrow$ audio